INT303 Coding Project2: Loan Approval Prediction

1 Introduction

In the financial field, the loans play a crucial role for economic growth and profit realization. Therefore, developing a machine learning model to realize accurate and efficient loan approval decisions is important. This project develops machine learning models to predict loan approval status based on applicant data including financial assets, credit scores, and other relevant features. We develop and compare three predictive models: Logistic Regression, Decision Tree, and Random Forest. In the end, we provide suggestions for applying the model based on actual situations and propose considerations for ethics.

2 Methodology

Firstly, we conducted systematic data preprocessing and feature engineering operations on the original dataset to improve model performance and training efficiency

1. **Missing Value Handling**: For numerical features, use median or mean to fill in. For categorical features, use mode to fill in.
2. **Categorical Variables Encoding**: Perform one hot encoding on categorical variables in the original dataset, including 'education', ' self_employed', ' loan_status', and convert them into numerical forms that the model can handle. The original categorical variables are encoded and removed from the dataset to avoid redundancy and collinearity issues.
3. **Feature Engineering**: To reduce dimensionality, improve model stability, and enhance interpretability, we use 'total_assets' to instead of features of different assets. We also add 'loan_div_income' to measure the repayment ability of customers.
4. **Normalization**: Considering that we will use models sensitive to feature scales such as logistic regression in the future, as well as accelerate model convergence and improve model generalization ability, we standardize the numerical features.
5. **Data Splitting**: Divide the processed data into training and testing sets in an 8:2 ratio.

3 Key Findings

Figure 1 shows the confusion matrix of decision tree and random forest in the project(these two model has the same performance). Based on the feature importance output by the two models, we have determined that they are not the same. We also know that 'cibil_score' is the most important feature, which has 0.82 importance.

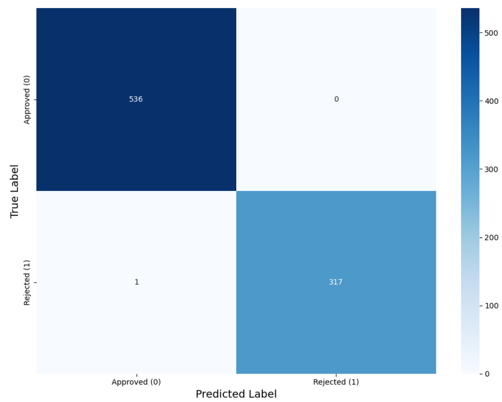

Figure 1: Confusion Matrix of Decision Tree and Random Forest

4 Recommendation & Insights

Therefore, we found that 'cibil_score' is the most critical factor in determining whether a loan is rejected. Other features are not so important.

Due to the same performance of decision tree and random forest, we provide different suggestions for different real-life situations. If the bank requires lower costs and higher interpretability, then decision trees are a better choice. If the bank wants higher stability and accuracy, random forest is better.

In the future, we need to collect more data to measure these two models, such as loan situations over more periods. Besides, the model can be tested under different train-test splits or with repeated cross-validation to confirm stability.

5 Ethical Considerations

Firstly, our data volume is actually not large enough, so the model we obtain may have deviations in practical applications. We need more data to evaluate the model.

Besides, Our model needs to be explained to the public in an interpretable way to avoid potential regulatory and trust issues.

Lastly, it is still recommended to use it in conjunction with manual labor to avoid accountability issues.